

# DIMPLE SAINI

Senior .NET Full Stack Developer | Cloud, Real-Time & AI-Integrated Systems

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Indian national based in India • Open to relocation to the UK • Requires Skilled Worker visa sponsorship

## PROFESSIONAL PROFILE

Senior .NET Full Stack Developer with 8+ years building and operating production enterprise systems for a national media organisation and the higher-education sector. Specialist in the full Microsoft stack (.NET Framework through .NET 9), Clean Architecture and Domain-Driven Design, RESTful APIs, real-time systems with SignalR, and secure authentication (JWT, OTP, RSA digital signing). Proven record of delivering platforms serving 2,000+ concurrent users, embedding LLM and AI services into live editorial pipelines, and leading a zero-downtime legacy modernisation against a live production database. Comfortable owning a system end to end — architecture, data modelling, security, CI/CD and production support.

## CAREER HIGHLIGHTS

- Migrated a legacy ASP.NET Web Forms platform to ASP.NET Core (.NET 8) against the live production database with zero downtime and no schema fork.
- Optimised SQL Server queries and schema design to sustain 2,000+ concurrent users on live, public-facing election and analytics dashboards.
- Introduced AI-driven news processing and automated video generation into a national newsroom, materially reducing manual editorial effort.

## CORE TECHNICAL SKILLS

**Languages & Frameworks:** C#, .NET Framework, .NET 8, .NET 9, ASP.NET Core, ASP.NET MVC, Razor Pages, Web API / REST, Entity Framework, Dapper, TypeScript, JavaScript

**Architecture:** Clean Architecture, Domain-Driven Design, Repository & Service Layer patterns, Dependency Injection, Microservices, API Gateway / reverse-proxy design, background & worker services

**Real-Time & Distributed:** SignalR, distributed collector/agent architectures, multi-service systems, stream-health monitoring

**Frontend:** React, TypeScript, Vite, HTML5, CSS3, jQuery, Bootstrap, AJAX

**AI & LLM Integration:** OpenAI API, multi-provider LLM orchestration (Anthropic Claude, Groq, Google Gemini, Mistral, OpenRouter, local inference via Ollama), agentic pipeline design, AI video-generation APIs, speech-to-text (Web Speech API, Google Speech Recognition)

**Databases:** Microsoft SQL Server, stored procedures, query optimisation, performance tuning, versioned schema migrations

**Security:** JWT and OTP authentication, RSA digital signing, BCrypt hashing, API-key auth, IP whitelisting, replay-attack protection, RBAC, device fingerprinting

**DevOps & Tooling:** GitHub Actions CI/CD, Git, IIS deployment and production server configuration, Postman, Swagger / OpenAPI

**Integrations:** YouTube Data API, Twitter/X API, SMS gateways, GPS APIs, RFID and hardware integration (RAID / BMC / SMART), WMI / Windows systems programming

## PROFESSIONAL EXPERIENCE

**Senior .NET Full Stack Developer | Zee Media Corporation Ltd**

August 2022 – Present | India | National broadcast media group

- Design and deliver real-time election-data automation systems powering live, public-facing broadcast dashboards under strict accuracy and uptime requirements.
- Built live social-media analytics dashboards integrating the YouTube and Twitter/X APIs, using a distributed collector plus stream-health service architecture for continuous monitoring.
- Developed AI-based news processing and content automation using the OpenAI API, integrated AI video-generation APIs into production media pipelines, and implemented speech-to-text and voice-recognition features for newsroom and teleprompter workflows.
- Optimised SQL Server queries and schema design, improving response times across systems supporting 2,000+ concurrent users.

- Architected an RFID warehouse inventory-tracking API (.NET 9, Clean Architecture) with replay-attack protection, API-key authentication, IP whitelisting and full audit logging.
- Built an enterprise hardware-monitoring platform on a distributed Collector → Server → API architecture, integrating RAID / BMC / SMART vendor tooling, 35+ versioned SQL migrations and interactive network-topology visualisation.
- Designed an internal application gateway and governance portal (.NET 9) with reverse-proxy middleware, JWT auth, device fingerprinting and a SignalR real-time dashboard, centralising access to internal enterprise tools.
- Led a zero-downtime migration of the legacy ASP.NET Web Forms Stringer Management System to ASP.NET Core (.NET 8) Razor Pages against the live production database, using an additive-only schema strategy with full audit trail and role-based dashboards (SuperAdmin / HR / Assignment / Editor).
- Established GitHub Actions CI/CD pipelines for automated build and deployment, and manage production applications on IIS including code signing and release pipelines.
- Delivered further internal enterprise systems — Exit Management (full-and-final settlement), Asset Management, an SMS automation platform, and an event pass-verification system secured with OTP authentication and RSA digital signing.

## **.NET Developer | Vardhman Infotech**

*October 2017 – December 2021 | India | Education technology*

- Led the digitisation of university result and academic-record systems, migrating legacy applications and paper-based processes onto modern .NET web platforms and improved coding standards.
- Developed academic certificate generation and online verification portals used across university stakeholder groups.
- Built student result-processing and entrance-exam automation systems handling large institutional datasets.
- Designed secure document management and acknowledgement systems for sensitive academic records.

## **INDEPENDENT PROJECTS**

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### **Live AI Fact-Checker Browser Extension (FakeCheck)** — *JavaScript, Manifest V3, multi-provider LLM*

Real-time extension that transcribes live speech (Web Speech API / AudioWorklet) and runs an agentic LLM pipeline to verify claims with confidence scores and sources. Pluggable architecture supporting six LLM providers, including local inference via Ollama. Cross-browser: Chrome, Edge, Brave, Opera and Firefox.

### **Real-Time 3D Racing Game (Nitro Rush)** — *Three.js, Rapier (Rust/WASM physics), Vite*

Browser-based arcade racer built on a WASM physics engine, featuring a raycast vehicle controller, drift and nitro mechanics, HDRI-based PBR rendering and post-processing — demonstrating real-time simulation and graphics-engine depth.

## **EDUCATION**

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### **Bachelor of Computer Applications (BCA)**

*Government P.G. College, Ambala Cantt, India | Graduated 2016*

## **ADDITIONAL INFORMATION**

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**Awards:** Appreciation Certificate, Kurukshetra University, for contribution to software development; State-Level Computer Science Competition winner (2014, 2015, 2016)

**Right to Work:** Indian national, currently living and working in India. No existing UK work authorisation — a Skilled Worker visa and employer sponsorship would be required. Open to relocation.

**Interests:** Cricket (former State-Level U-19 player), open-source development, real-time graphics and simulation

**References:** Available on request